

The Type I C^* -Algebra Ball-Covering Classification

Run `fa_banach_001`, lane 19

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Source Problem

The source paper is Jinghao Huang, Karimbergen Kudaybergenov, and Rui Liu, *The ball-covering property of von Neumann algebras and noncommutative symmetric spaces*, arXiv:2311.14257. After proving that a commutative unital C^* -algebra $C(K)$ has BCP/UBCP exactly when K has a countable π -basis, the authors ask for a noncommutative version:

$\{U_\alpha\}_{\alpha \in A}$ of K is called a π -basis (weak basis) for K if every nonempty open subset of K contains some U_α in $\mathcal{U}$ [37, Page 15].

In the setting of commutative C^* -algebra, [25, Theorem 2.1] can be reformulated as follows. Note that [25, Theorem 2.1] only deals with real continuous functions. The complex case is a direct consequence of the real case.

Corollary 3.9. *Let $\mathcal{A}$ be a unital commutative C^* -algebra. Then the followings are equivalent:*

- (1) $\mathcal{A}$ has the BCP;
- (2) the pure state space $\mathfrak{P}(\mathcal{A})$ has a countable π -basis;
- (3) $\mathcal{A}$ has the UBCP.

In particular, there exist commutative C^* -algebras containing no minimal projections but having the BCP. For example, the uniform norm closure of the linear span of Haar functions [24, p.150].

We conclude this section by the following question.

Question 3. *It will be interesting to find a non-commutative version of Corollary 3.9, that is, how to characterize those C^* -algebras $\mathcal{A}$ having BCP/UBCP.*

Definitions

A normed space X has the ball-covering property, or BCP, if its unit sphere is covered by countably many open balls none of which contains the origin. Equivalently, X has BCP if and only if there are countably many centres $x_n \in X$ such that

$$S_X \subseteq \bigcup_n B(x_n, \|x_n\|).$$

It has UBCP if the balls can be chosen with a common positive gap from the origin.

For an open set $O \subseteq \text{Prim}(A)$, let A_O denote the corresponding closed two-sided ideal. For a Type I algebra A , write $\text{Fell}(A)$ for the largest Fell ideal.

Statement

Theorem 1 (nonseparable elementary open patches obstruct BCP). *Let A be a C^* -algebra. Suppose there is a nonempty open set $O \subseteq \text{Prim}(A)$ such that*

$$A_O \cong C_0(O, \mathcal{K}(H))$$

for a nonseparable Hilbert space H . Then A fails BCP.

Theorem 2 (complete Type I criterion). *Let A be a Type I C^* -algebra. Then the following are equivalent:*

- (i) A has BCP;
- (ii) A has UBCP;
- (iii) $\text{Prim}(A)$ has a countable π -basis and every irreducible representation of $\text{Fell}(A)$ is separable.

Idea of the Proof

The positive half was already proved in the essential-Fell packet: a Type I algebra has an essential largest Fell ideal, and if the irreducible representations on that Fell ideal are separable, countably many atomic multiplier charts cover the sphere uniformly.

The missing half is to show that nonseparable Fell fibres really are visible to BCP. The key is local. On a trivial patch

$$C_0(O, \mathcal{K}(H)) \triangleleft A$$

with H nonseparable, choose a scalar bump h with compact support $S \subset O$. A countable list of centres acts on this ideal as bounded strict-continuous $B(H)$ -fields $T_n(P)$. Over the compact set S , strict continuity makes each orbit $P \mapsto T_n(P)\xi$ norm-compact. Therefore one can build a separable subspace $L \subset H$ reducing all operators $T_n(P)$, $P \in S$, while still norming each field on S .

Put a rank-one projection R in $L^\perp$, and set

$$p(P) = h(P)R.$$

On S , the difference $p(P) - T_n(P)$ has the untouched block $-T_n(P)|_L$. Outside S , p vanishes. Thus p is a unit-sphere element escaping all BCP balls.

The Local Obstruction

Lemma 1 (separable reducing subspace for compact strict families). *Let S be a compact Hausdorff space, let H be a Hilbert space, and for each $n \geq 1$ let*

$$P \mapsto T_n(P) \in \mathcal{B}(H)$$

be a bounded field such that both $P \mapsto T_n(P)\xi$ and $P \mapsto T_n(P)^\xi$ are norm-continuous for every $\xi \in H$. Then there is a separable closed subspace $L \subseteq H$ such that:*

1. L reduces every $T_n(P)$, for all n and $P \in S$;
- 2.

$$\sup_{P \in S} \|T_n(P)|_L\| = \sup_{P \in S} \|T_n(P)\| \quad (n \geq 1).$$

Proof. Put

$$\alpha_n = \sup_{P \in S} \|T_n(P)\|.$$

For each n, k , choose $P_{n,k} \in S$ and a unit vector $\xi_{n,k} \in H$ such that

$$\|T_n(P_{n,k})\xi_{n,k}\| > \alpha_n - 1/k$$

when $\alpha_n > 0$; if $\alpha_n = 0$, choose arbitrary unit vectors.

We construct increasing separable subspaces L_m . Let L_0 be the closed span of the countable set $\{\xi_{n,k} : n, k \geq 1\}$. Given L_m , choose a countable dense subset D_m of its unit ball. For each $\xi \in D_m$ and each n , the sets

$$\{T_n(P)\xi : P \in S\}, \quad \{T_n(P)^*\xi : P \in S\}$$

are compact metric subsets of H , hence separable. Let L_{m+1} be the closed span of L_m together with countable dense subsets of all these compact image sets, over all n and $\xi \in D_m$.

Set

$$L = \overline{\bigcup_{m=0}^{\infty} L_m}.$$

Then L is separable. The construction and boundedness of the fields show that $T_n(P)L \subseteq L$ and $T_n(P)^*L \subseteq L$ for every n, P : first this holds on each dense set D_m , then by norm approximation on L_m , and finally by closure. Hence L reduces every $T_n(P)$.

Since each $\xi_{n,k}$ lies in L ,

$$\sup_{P \in S} \|T_n(P)|_L\| \geq \sup_k \|T_n(P_{n,k})\xi_{n,k}\| = \alpha_n.$$

The reverse inequality is automatic, so equality holds. $\square$

Proof of Theorem 1. Identify A_O with $C_0(O, \mathcal{K}(H))$. Let (a_n) be any countable family of centres in A . Choose $h \in C_c(O)$ with

$$0 \leq h \leq 1, \quad \|h\| = 1,$$

and put $S = \text{supp } h$, a nonempty compact subset of O .

The action of A on the ideal A_O gives, for each a_n , a bounded strict-continuous multiplier field

$$P \mapsto T_n(P) \in \mathcal{B}(H) \quad (P \in O).$$

On the compact set S , these fields satisfy the hypotheses of Lemma 1. Choose the corresponding separable reducing subspace L . Since H is nonseparable, $L^\perp \neq \{0\}$. Pick a unit vector $\eta \in L^\perp$, let R be the rank-one projection onto $\mathbb{C}\eta$, and define

$$p(P) = h(P)R \quad (P \in O).$$

Then $p \in C_c(O, \mathcal{K}(H)) \subseteq A_O \subseteq A$, and $\|p\| = 1$.

Fix n . Put

$$\alpha_n = \sup_{P \in S} \|T_n(P)\|, \quad \beta_n = \sup_{P \in \text{Prim}(A) \setminus S} \|a_n + P\|.$$

Here, on O , $\|a_n + P\| = \|T_n(P)\|$. Since the C^* -norm is the supremum over primitive quotients,

$$\|a_n\| = \max\{\alpha_n, \beta_n\}.$$

For $P \notin S$, the element p vanishes in the primitive quotient at P , so

$$\|p + P - (a_n + P)\| = \|a_n + P\|.$$

Thus the contribution from $\text{Prim}(A) \setminus S$ is at least β_n .

For $P \in S$, the subspace L reduces $T_n(P)$, while R acts on $L^\perp$. Therefore

$$h(P)R - T_n(P) = (-T_n(P)|_L) \oplus (h(P)R - T_n(P)|_{L^\perp})$$

relative to $H = L \oplus L^\perp$. Hence

$$\|h(P)R - T_n(P)\| \geq \|T_n(P)|_L\|.$$

Taking the supremum over $P \in S$ and using Lemma 1 gives a contribution at least α_n . Combining the two regions,

$$\|p - a_n\| \geq \max\{\alpha_n, \beta_n\} = \|a_n\|.$$

Thus the unit-sphere element p is outside every open ball $B(a_n, \|a_n\|)$. Since the countable family of centres was arbitrary, A fails BCP. $\square$

The Type I Classification

Proof of Theorem 2. First suppose A has BCP. The primitive π -base obstruction from the earlier lane-19 packet gives a countable π -basis for $\text{Prim}(A)$. We claim that every irreducible representation of $\text{Fell}(A)$ is separable. If not, choose a nonseparable irreducible representation in the Fell ideal. Fell spectra are locally Hausdorff, and on Hausdorff open patches Fell algebras are continuous trace [2]. Local triviality of continuous trace algebras gives a nonempty open set

$$O \subseteq \text{Prim}(\text{Fell}(A)) \subseteq \text{Prim}(A)$$

such that

$$A_O = \text{Fell}(A)_O \cong C_0(O, \mathcal{K}(H))$$

for a nonseparable Hilbert space H . Theorem 1 would then force A to fail BCP, a contradiction. Thus condition (iii) is necessary.

The implication $\text{UBCP} \Rightarrow \text{BCP}$ is immediate.

Conversely, assume (iii). The previous Type I packet proved that every Type I algebra has an essential largest Fell ideal $\text{Fell}(A)$. Since all irreducible representations of $\text{Fell}(A)$ are separable, the essential-Fell theorem [5] applies: the countable π -basis of $\text{Prim}(A)$ refines to countably many π -dense trivial continuous-trace patches with separable fibres, and the atomic multiplier chart construction gives UBCP for A . Hence (iii) implies (ii).

The three conditions are therefore equivalent. $\square$

Scope and Limitations

This packet gives a complete answer for Type I C^* -algebras. It also subsumes the one-point nonseparable compact-operator obstruction: if $\mathcal{K}(H) \subseteq A \subseteq \mathcal{B}(H)$ and H is nonseparable, then $\text{Prim}(A)$ has an open elementary nonseparable patch, so A fails BCP.

The arbitrary C^* -classification remains open. Outside Type I, separable non-Type-I algebras still have UBCP, while nonseparable non-Type-I algebras can fail by atomless von Neumann or diagonal quotient mechanisms. The remaining problem is to determine exactly when those non-Type-I mechanisms are visible to the ambient C^* -norm.

Novelty and Search Bounds

The local indexes were searched for arXiv:2311.14257, Type I, Fell ideal, continuous trace, nonseparable compact operators, intermediate algebras $\mathcal{K}(H) \subseteq A \subseteq \mathcal{B}(H)$, BCP, UBCP, and primitive π -bases. Existing packets included the essential-Fell positive theorem, the Type I separable-irrep theorem, and the intermediate nonseparable compact obstruction, but no complete Type I iff classification.

Web searches on June 17, 2026 for intermediate $K(H)/B(H)$ algebras, nonseparable compact operators, and BCP located nearby Calkin/operator-space material but no exact theorem of the above form. The Fell and continuous-trace structure input is standard and cited via Clark–an Huef–Raeburn, Dixmier, and Raeburn–Williams.

Human Review Focus

Reviewers should check:

1. Lemma 1, especially the use of strict continuity to keep the reducing subspace separable over a compact support;
2. the split between the compact support S and its complement in the proof of Theorem 1;
3. the passage from a nonseparable irreducible representation of $\text{Fell}(A)$ to an open trivial continuous-trace patch;
4. the handoff to the earlier essential-Fell positive theorem.

References

- [1] Jinghao Huang, Karimbergen Kudaybergenov, and Rui Liu, *The ball-covering property of von Neumann algebras and noncommutative symmetric spaces*, arXiv:2311.14257, 2023.
- [2] Lisa Orloff Clark, Astrid an Huef, and Iain Raeburn, *The equivalence relations of local homeomorphisms and Fell algebras*, arXiv:1207.2540, 2012.
- [3] J. Dixmier, *C^* -Algebras*, North-Holland, 1977.
- [4] I. Raeburn and D. P. Williams, *Morita Equivalence and Continuous-Trace C^* -Algebras*, American Mathematical Society, 1998.
- [5] Run `fa_banach_001`, lane 19, *Essential Fell Ideals and the Ball-Covering Property*, solution packet `2311.14257_essential_fell_ideal_bcp`, 2026.
- [6] Run `fa_banach_001`, lane 19, *Type I C^* -Algebras and the Ball-Covering Property*, solution packet `2311.14257_type_i_separable_irrep_bcp`, 2026.
- [7] Run `fa_banach_001`, lane 19, *Intermediate Algebras over Nonseparable Compacts Fail BCP*, solution packet `2311.14257_intermediate_nonseparable_compacts_fail_bcp`, 2026.